shown in Table 412-C.

412.33.02 Bar Supports and Spacers - Bar supports and spacers shall be adequate to ensure concrete cover and bar spacings are maintained within the specified tolerances.

Bar supports and spacers shall be sufficient in number and strength to support the reinforcement and prevent displacement by workers or equipment before and during concreting and shall be adequately spaced to ensure that any sagging between supports does not intrude on the specified concrete cover.

TABLE 412-C TOLERANCES FOR PLACING REINFORCEMENT

Concrete cover to top reinforcing steel in bridge decks	+ 6 mm - 0 mm
Concrete cover to reinforcement, other than top reinforcing steel in bridge decks	± 8 mm
Bar location, except cover, when depth of a flexural member, thickness of a wall or smallest dimension of a column is 200 mm or less	± 8 mm
Bar location, except cover, when depth of a flexural member, thickness of a wall or smallest dimension of a column is larger than 200 mm but less than 600 mm	± 12 mm
Bar location, except cover, when depth of a flexural member, thickness of a wall or smallest dimension of a column is 600 mm or larger	± 20 mm
Longitudinal location, except cover, of bends and ends of bars	± 50 mm
Longitudinal location, except cover, of bends and ends of bars at discontinuous ends of members	± 20 mm

Bar supports and spacers shall be of a type and material that will not cause rust spots, blemishes or spalling of concrete surfaces.

Bar supports and spacers shall be precast concrete, plastic, steel wire, stainless steel or steel bar, except that, supports or spacers over 200 mm in height shall be precast concrete or steel bar.

Bright wire and uncoated steel supports and spacers may only be used where they are not in contact with soil surfaces, finished concrete surfaces or epoxy coated reinforcement.

Where concrete surfaces are to be exposed to sandblasting, sea water, or de-icing chemicals the bar supports and spacers shall be either stainless steel, hot dipped galvanized steel, epoxy coated reinforcing steel, plastic or precast concrete.

Precast concrete supports shall have a compressive strength and quality not less than that of the concrete in which they are embedded. For finished surfaces, the face of the support in contact with the forms shall not exceed 50 mm in any dimension and shall have a colour and texture to match that of the finished concrete surface.

Bar supports and spacers for approach slabs and formed horizontal slabs such as bridge decks, bridge sidewalks, and top slabs of culverts shall meet the following additional requirements:

a) Supports or spacers up to 200 mm in height shall be either precast concrete or plastic.

b) Supports and spacers over 200 mm in height shall be either:

i) precast concrete,

ii) bent or welded steel bar that has been hot-dipped galvanized after fabrication,

iii) stainless steel bar, or

iv) epoxy coated reinforcing steel bar

c) Plastic supports or spacers shall have an individual minimum breaking strength of 4.5 kN (1000 lbs) and shall be capable of maintaining strength and dimensional properties for the range of temperatures encountered on site.

d) Maximum spacing between supports, for each mat of bars, shall be 1200 mm, except that the maximum spacing between plastic supports, for each mat of bars, shall be 1000 mm.

e) Each support shall carry the load from not more than one mat of reinforcing steel, except as provided below:

i) Where a support is used that is specifically designed by the manufacturer to carry two mats at two separate positions, or

ii) If the top mat of reinforcing steel is more than 200 mm above the slab soffit, then the top mat may be supported by epoxy coated reinforcing steel spreader bars secured between the top and bottom reinforcing mats, provided that, the bottom reinforcing mat is supported on precast concrete chairs which are sufficient in number and adequately spaced to carry the additional weight of the top mat of reinforcing steel.

TABLE 412-D CONCRETE COVER FOR REINFORCING STEEL

Concrete cast against ground	75 mm
Tops of deck slabs	70 mm
Undersides of deck slabs	40 mm
Other surfaces	60 mm

412.33.03 Concrete Cover - Concrete cover for reinforcing steel shall be as listed in Table 412-D, unless shown otherwise on the Design.

412.33.04 Spirals - Spirals shall be held in place and to line by vertical spacers.

For spiral rods less than 15M spacing shall be as follows:

- two spacers per loop for spirals less than 500 mm in diameter.
- three spacers per loop for spirals 500 mm to 800 mm in diameter.
- four spacers per loop for spirals over 800 mm in diameter.

For spiral rods 15M and larger spacing shall be as follows:

- three spacers per loop for spirals up to 600 mm in diameter.
- four spacers per loop for spirals over 600 mm in diameter.

If vertical reinforcements are to serve as spacers, spirals shall be securely tied to vertical reinforcement at the spacing required for spacers.

412.34 Splicing of Bars - Special requirements for splicing, such as particular locations for splices, use of overlength bars or special lap lengths, shall be as shown on the Design.

Splices in bars larger than 35M shall be mechanical coupler splices or welded. The detail of such splices shall be subject to acceptance by the Designer and Ministry Representative.

412.34.01 Lapped Splices

a) Horizontal bars shall not be spliced unless greater than standard mill lengths - 12 m for 10M bars and 18 m for larger bars - would otherwise be required.

b) Splices to be staggered so that no more than one third of the reinforcing steel in a member shall be spliced in any transverse section that is within the required lap length.

c) If not shown on the Design, the length of lapped splices shall be in accordance with CAN/CSA-S6-00, "Canadian Highway Bridge Design Code", Clause 8.15.9.

d) Adjacent reinforcing bars shall not be spliced at the same locations unless shown on the Design or authorized by the Designer and Ministry Representative.

Column spirals shall be lapped 48 rod diameters at splices unless otherwise shown on the Design.

412.34.02 Welded Splices - Welded splices shall be used only if detailed on the Design. Welding shall conform to the requirements of CSA Standard W186, Welding of Reinforcing Bars in Reinforced Concrete Construction and shall be performed by a company certified by the Canadian Welding Bureau to the requirements of CSA Standard W186. Welded splices shall not be used on epoxy coated bars.

412.34.03 Mechanical Coupler Splices - Mechanical couplers shall be used for splices only if pre-approved or detailed on the Design. Such couplers shall develop in tension or compression, as required, at least 120% of the specific yield strength of the bars, but not less than 110% of the mean yield strength, representative of the bars to be used, in the test of the mechanical connection.

Wedge couplers shall not be used in bars greater than 15 mm diameter.

When mechanical couplers are used the Contractor shall ensure that the minimum concrete cover as specified is maintained.

The total slip of the reinforcing bars within the splice sleeve of the connector after loading in tension to $0.5f_y$ and relaxing it to $0.05f_y$ shall not exceed the following measured displacements between gauge points straddling the splice sleeve:

Up to and including 45M bars	0.25 mm
55M bars	0.75 mm

412.81 Not Used

412.91 Not Used

SECTION 413

BRIDGE DECKS AND CONCRETE OVERLAYS

DESCRIPTION

413.01 General - This Section describes requirements for the construction of Bridge Decks, and for the construction of Concrete Overlays on previously constructed (old) bridge decks and on newly constructed (new) bridge decks. The requirements of DBSS 211 shall apply except as otherwise specified in this Section. Unless otherwise specified in this Section, bridge deck concrete shall include diaphragms, deck slabs, bridge parapets and medians, bridge sidewalks, approach slabs and overlay concrete.

413.31 Bridge Decks

413.31.01 General - Deck slab construction shall be full-width and continuous between the joints shown on the Design, and shall be placed in the sequence as shown on the Design. Any variation from this requirement shall be to the acceptance of the Designer and the Ministry Representative.

413.31.01.01 Camber Survey - Prior to commencing deck construction and subsequent to the completion of the erection of girders, with the girders in a thermally neutral condition and with no dead load other than the dead load of the girders and necessary fall protection, the Contractor shall profile all the girders. . Camber elevations are to be taken at 1000 mm centres maximum (or as indicated on the Design), centrelines of all bearings and ends of girders. The camber elevation locations are to be laid out to an accuracy of 25 mm horizontally and camber elevations taken to the nearest 1 mm vertically. The Contractor shall supply the camber elevations to the Designer. The Designer shall calculate and provide the Contractor with the haunch heights for setting the deck slab soffit, and the design finished grade elevations for the deck, complete with anticipated dead load deflections at 1000 mm stations along the centreline of roadway. The Designer will be responsible for providing the elevations to be used by the Contractor in establishing the required screed rail settings.

In the event that actual girder camber values vary significantly from the design values, the Contractor will be required to modify the original design finished grade as directed by the Designer.

413.31.01.02 Method, Equipment & Personnel - The Contractor shall submit to the Quality Manager and the Ministry Representative for review, four weeks prior to the scheduled start of deck slab concrete placement, a Deck Placement Plan which provides a complete description of the method, equipment and personnel, including previous deck placement experience of the personnel to be employed. This shall include the equipment for mixing, transporting, placing, vibrating and curing the concrete and shall also include details of the finishing equipment and support rails. The submittal shall clearly demonstrate that the proposed methods, equipment and personnel have the capability to carry out the Work in accordance with DBSS 413. All review concerns raised by the Quality Manager and the Ministry Representative shall be addressed to the satisfaction of the Ministry Representative. Equipment proposed for the Work shall be subject to review by the Ministry Representative. Equipment may be rejected during operation if it is unsatisfactory.

Two weeks prior to the scheduled start of deck slab concrete placement, the Contractor shall meet with the Quality Manager, concrete supplier and the Ministry Representative to review deck concrete operations with respect to concrete mix design and performance, concrete delivery, placing, finishing and curing equipment and methods.

413.31.01.03 Haunch Setting, Screed Rails & Reinforcing Cover – The Contractor shall set the deck slab soffit formwork to within 2 mm of the haunch dimension as determined in accordance with DBSS 413.31.01.01.

After installation of the deck slab reinforcing, the Contractor shall set the screed rails for the deck finishing machine. Screed rails shall be installed to an accuracy of ± 2 mm and have a maximum deflection between supports of 1 mm. Allowance for dead load deflection of the structure shall be made in the screed rail setting as well as any anticipated deflection due to the loading of any formwork that may be supporting the screed rail system. The screed rails shall be adequately supported outside the area to be concreted and shall be completely installed before concrete is placed.

Prior to the placement of deck slab concrete and subsequent to the installation of deck slab reinforcing and screed rails, the locations of the top reinforcing steel in bridge decks shall be checked by running a full deck-width template along the longitudinal screeds. The lower edge of the template shall be set at the nominal cover dimension (as per the Design) plus any anticipated deflection of the screed rail support system below the level of the deck surface. No steel shall touch the template nor be more than 6 mm distant from it. Any discrepancies outside of these limits shall be

corrected by the Contractor to the satisfaction of the Designer and Ministry Representative.

413.31.02 Concrete Placement

413.31.02 .01 Concrete Temperature at Placement - Concrete designed for bridge decks including any bridge deck concrete containing silica fume shall not have a temperature in excess of 17°C at time of placement. Concrete used in elements having a minimum section thickness greater than 1000 mm shall have temperature controls as per CSA A23.1 Clause 5.2.4.4. Methods used for maintaining temperature controls shall not diminish the quality of the concrete, and shall not exceed the specified W/C_m ratio.

When the atmospheric temperature is 15°C or higher, DBSS 211.18 – Hot Weather Concreting, shall apply except as modified herein.

Concrete for bridge decks and overlays shall not have a temperature greater than 17°C when placed. Prior to the placement of bridge deck and overlay concrete, when the ambient air temperature is at or above 15 °C, or when there is a probability of its rising to 20°C or above during the placement period (as forecast by the nearest official meteorological office), the Contractor shall submit to the Quality Manager and Ministry Representative for review, the proposed placing operations complying with the requirements for hot weather concreting.

When the ambient air temperature is at or above 15°C, or when there is a probability of it's rising to 20°C or above during the placement period (as forecast by the nearest official meteorological office), bridge decks and overlays shall be cast when daily air temperatures are forecast to be a minimum (normally between 12:00 midnight and 9:00 AM).

413.31.02 .02 Time of Placement – All bridge deck and overlay concrete shall be fully discharged and placed in its final location within 45 minutes after the water and cement have been combined. Under conditions contributing to rapid stiffening of concrete, the Contractor shall adjust its placing operations to suit the reduced handling time. .

Time of placement extensions will be considered on a case by case basis and may require the use of Hydration Stabilizing Admixtures (HSA's) conforming to the requirements of ASTM C494 Type B, Retarding or Type D, Water-Reducing and Retarding Admixtures. If accepted by the Designer and the Ministry Representative, guidelines for the use of HSA's shall be as follows:

- HSA modified concrete for bridge decks shall be fully discharged and placed within 90 minutes after water and cement have been combined.

- When HSA's are used, these time extensions are subject to preconstruction trials being conducted by the Contractor, to establish the appropriate HSA dosage to provide suitable extended slump life of concrete, without increasing the water/cementitious ratio of the concrete above that which would be required if HSA's were not used. The use of HSA's shall in no instance modify the maximum concrete temperature required at time of placement. A one time only addition of HSA will be allowed. This will be during initial batching of the concrete or immediately at completion of batching as recommended by the admixture manufacturer. Addition of HSA at any other time will be cause for rejection of the concrete.

413.31.02 .03 Concrete Placement-Diaphragms - Unless otherwise specified by the Designer, diaphragms shall be placed either:

a) about two hours before the deck concrete. In this case, diaphragm concrete shall have a slump less than 40 mm and shall be retarded so that it sets at the same time as the deck concrete, or

b) a sufficient length of time ahead of the deck concrete, so that the diaphragm concrete attains a compressive strength of 15 MPa before the deck slab is cast.

Sequence for the placing of concrete for diaphragms at ends of girders on continuous and integral type spans shall be as shown on Design.

413.31.02 .04 Concrete Placement-Deck Slab - The top surfaces of concrete stringers shall be prepared in accordance with DBSS.211.12.03 and the surfaces shall be in a saturated surface dry condition immediately before the deck is cast.

Concrete shall be placed at a rate of not less than 6 m/hr. The concrete front and finishing operation shall be kept parallel to the substructure. Placing shall be started at such time as to permit finishing during daylight hours, except if ambient temperatures are as outlined in DBSS 413.31.02.01, then adequate lighting is to be provided in accordance with Workers Compensation Act, Occupational Health and Safety Regulations, BC, to allow for concrete operations under reduced light conditions.

In the event of unexpected rain, concrete placing shall cease and the surface shall be protected immediately.

Surface concrete which has been rained on shall immediately be removed to the top of reinforcing steel.

413.31.02 .05 Finishing - The deck surface shall be finished by a drum-type deck finisher on support rails. Two work bridges shall be provided. No inserts will be permitted in the finished roadway surface.

The surface behind the finisher shall be floated transversely to eliminate ridges and open texture. Areas which cannot be machine-finished shall be hand screeded and floated to conform to the machine-finished deck.

Prior to final surface texturing, the deck surfaces shall be repeatedly checked for accuracy of deck slab thickness and alignment. The Contractor shall measure and record the deck slab thickness at a frequency of at least one measurement per interior bay and overhang bay, at a maximum of 3000 mm centers longitudinally. Adjustments as acceptable to the Designer shall be made to the deck finishing machine to maintain the minimum deck slab thickness and the required concrete cover on the deck reinforcing. The deck slab thickness readings shall be supplied to the Quality Manager and upon request, the Ministry Representative at the conclusion of each placement section.

Bridge deck slab final surface texturing shall be by tining. The tining shall create transverse grooves 3 mm wide by 1.5 mm to 3 mm deep at 20 mm centre to centre spacing.

413.31.03 Repairs - Areas of the deck not meeting the required accuracy of alignment as given in DBSS 211.16 Table 211-M, shall be corrected as follows:

a) where the required alignment can be obtained with a maximum cut of 6 mm, the bumps shall be cut with a bump cutter which will provide an acceptable longitudinal grooved texture;

b) where the required alignment cannot be obtained with a maximum cut of 6 mm, the concrete shall be broken out with vertical edges to a minimum depth of 40 mm and shall be replaced with new concrete of the required quality.

Any physical damage to the deck surface shall be cut out and replaced. Any areas of open texture or plucked aggregates shall be repaired to the satisfaction of the Quality Manager and the Ministry Representative.

413.31.04 Interim Strength Requirements - Deck concrete shall attain a strength of 15 MPa before parapets are placed and 25 MPa before heavy loads, such as concrete trucks are allowed on the bridge.

413.31.05 Curing

413.31.05.01 General - Freshly deposited concrete shall be protected from freezing, abnormally high temperatures or temperature differentials, premature drying, excessive moisture, moisture loss, heavy shocks, excessive vibrations and high stresses, for the period of time necessary to develop the required concrete properties.

The Contractor shall submit a detailed Deck Curing Plan for curing deck and overlay concrete. The plan shall contain details such as water source, the distribution system, the collection and run-off control system, person(s) responsible for quality control, and staff to maintain the system. The curing system shall be capable of maintaining free film of water, on the areas of concrete requiring curing, for a minimum of 8 unattended hours and this capability shall be demonstrated prior to concrete placement. It shall be the responsibility of the Contractor to ensure that the system of curing and protection is properly planned, constructed and maintained throughout the entire curing period.

Burlap shall be pre-soaked by immersing it in water for a period of at least 24 hours immediately prior to placing. A layer of burlap shall be applied to the surface of the concrete. Strips must overlap 150 mm and must be held in place without marring the surface of the concrete.

The burlap shall be applied immediately after finishing of the concrete surface within 2 to 4 meters of the final surface texturing operation (no more than 20 minutes) and shall be maintained in a saturated condition in a manner which does not damage the finished concrete surface. The burlap shall be maintained in a continuously wet condition throughout the curing period and a continuous film of free water on the surfaces of the areas noted herein shall be maintained by means of soaker hoses, sprinklers, etc.

Concrete for diaphragms, monolithic decks, concrete overlays, approach slabs, parapets, and sidewalks shall be cured for a minimum of seven consecutive uninterrupted twenty-four hour periods at a minimum temperature of 10°C and for the time necessary to attain 80% of the specified compressive strength of the concrete. The compressive strength tests for verifying completion of the curing period shall be determined using field cured test cylinders cured in accordance with Clause 7.3.4 of CSA A23.2-3C. The curing period will begin following the completed placement of the concrete in the section cast.

The temperature of the concrete being cured must be monitored to ensure maximum and minimum temperature requirements in accordance with DBSS 211.15.02 are met.

Regardless of ambient temperature, moist curing with burlap and water must be provided at all times. During cold weather, burlap must be prevented from freezing.

At completion of the curing period, the curing shall be discontinued in a manner so as not to subject the concrete to undue stresses due to rapid moisture loss in the concrete.

When the air temperature is at or below 5°C or when there is a probability of it falling below 5°C within the next 5 days after the curing period, then the following shall apply:

- At completion of the curing period, the addition of free water is to be discontinued and followed by an additional 5 consecutive uninterrupted twenty-four hour periods of drying with the concrete temperature maintained at a minimum temperature of 5°C.

413.31.05.02 Additional Curing Requirements for Concrete Containing Silica Fume or Fly Ash, With or Without Fibres. - A fog mist shall be applied continuously from the time of screeding until the concrete is covered with burlap, in such a way as to maintain high relative humidity above the concrete and prevent drying of the concrete surface. Water must not be allowed to drip, flow, or puddle on the concrete surface during fog misting, when placing the burlap, or at any time before the concrete has achieved final set.

Fogging equipment will consist of pressure washers (minimum output pressure of 28 MPa (4000 psi)) with suitable tips for producing a fine water mist or fog. A minimum of two pressure washers shall be used for this process and the adequacy of the range and coverage shall be demonstrated prior to placing any concrete.

When the air temperature is at or below 5°C or when there is a probability of it falling below 5°C within the next 10 days after the curing period, then the following shall apply:

- At completion of the curing period, the addition of free water is to be discontinued and followed by additional 10 consecutive uninterrupted twenty-four hour periods of drying with the concrete temperature maintained at a minimum temperature of 10°C.

Modifications to the curing process may be required if it is anticipated that the deck may be subjected to the application of de-icing chemicals within the first month after the curing period.

CONCRETE OVERLAYS

413.32.01 General - This Section describes requirements for repair and resurfacing of existing bridge decks (old) and newly constructed bridge decks (new) with a concrete overlay, describing the construction practices, materials and equipment required to place acceptable concrete overlay. This work is usually carried out on one half of the deck at a time, while traffic is maintained on the other half. The requirements of DBSS 413.31 shall apply except as otherwise specified in this section.

413.32.02 Gradeline - The Contractor shall survey and reference profiles of the existing bridge deck and approaches prior to scarification as follows:

- Longitudinally the profiles shall be taken at 2 to 3 metre stations for the entire length of the structure, including 21 meters on each end, with additional shots taken at the back of ballast walls and all existing or proposed new joint locations.
- Transversely the number of profiles required will be determined by:
  - A maximum spacing between elevation points of 3 meters.
  - Profiles are required 500 mm from each curb or parapet face.
  - Profiles are required 300 mm (inside area to be concreted) from longitudinal overlay construction joint lines requiring adjustable screed rails. In general longitudinal overlay construction joints shall be located at the center of a lane or between lanes. The final location of the longitudinal construction joints is subject to the acceptance of the Designer and Ministry Representative.
  - A profile is required at hinge point of crown.
- Elevation points are to be laid out in cross section ninety degrees to centerline of roadway on straight structures and radial to centerline of roadway on curved or spiraled structures.
- Profile elevation points shall be laid out to an accuracy of ± 20 mm horizontally and elevations taken to an accuracy of ± 3 mm vertically and be referenced such that they can be re-established within ± 25 mm from their original location.

The Contractor shall provide the Designer, and when requested, the Ministry Representative, with the gradeline profiles plotted at a scale of 1:10 vertical and

1:50 horizontal and a spreadsheet showing actual original elevations for all points surveyed.
In the case of old decks, subsequent to scarifying and cleaning of the deck surface, the Contractor shall re-establish the original elevation points, take new elevations for the points to the nearest millimeter and submit these elevations along with plotted profiles to the Designer and , when requested, to the Ministry Representative.

The Designer will provide the Contractor with design finished grade profiles and elevations for the survey points as originally laid out by the Contractor. These profiles and elevations shall be used by the Contractor to set the screed rails for the deck finishing machine, checking of overlay thickness during the dry run of the deck machine and for calculating the volume of overlay concrete required.

The Contractor is responsible for properly setting the screed rails to match the design gradeline provided by the Designer. Depressions creating ponded water, or localized high spots in the concrete surface, resulting from deficient finishing procedures shall be repaired.

At completion of the deck overlay the Contractor shall re-establish the original elevation points and take as built elevations for the points to the nearest millimetre and submit these elevations along with plotted profiles to the Designer and Ministry Representative.

413.32.03 Removal of Existing Concrete Deck Surface

The existing concrete surface shall be removed by use of scarification (roto-milling), hydro-demolition or alternate methods acceptable to the Designer and Ministry Representative over the full length and width of the bridge deck. If methods other than scarification are proposed by the Contractor, the Contractor shall submit all required details of the proposed work to the Designer and Ministry Representative and shall obtain the acceptance of the Designer and Ministry Representative at least two weeks prior to starting the work. Alternate methods shall meet the requirements as specified for scarification.

413.32.03.01 Scarification - If scarified, equipment shall be capable of removal to an accuracy of ± 5 mm. The scarifying equipment shall produce a surface that does not exceed 10 mm in amplitude for roughness. The maximum cut will be to within 10 mm of the reinforcing steel. Depth of scarification will be specified by the Designer.

The Contractor shall monitor the total depth of concrete cover to the top mat of reinforcing steel using a pachometer or alternate methods acceptable to the Quality Manager. Readings shall be taken at regular intervals ahead of the scarification. The Contractor shall adjust the rate and depth of concrete removal to ensure that the reinforcing steel is not damaged and the desired scarified profile is achieved. All reinforcing steel damaged by the Contractor's operations will be replaced. Care must be taken not to damage the existing deck drains. The Contractor will be required to repair or replace any damaged drains. Drains are to be plugged to insure containment of debris.

All areas inaccessible to the scarification equipment shall be prepared by hydro-demolition, grinders or by jackhammers that are limited to a maximum 18 kg. Preparation shall minimize damage to the sound substrate concrete. Larger jackhammers may be used if accepted by the Designer and the Ministry Representative and if performance acceptable to the Designer and the Ministry Representative can be demonstrated.

In the case of previously overlaid decks, concrete shall be removed to below the old bond line. Where the bond line is within 10 mm of the reinforcing steel, the remaining concrete and the old bonding material shall be removed by hydro-demolition or pneumatic tools.

Upon completion of scarification of the deck, prior to opening the scarified deck lane to traffic, the Contractor shall place temporary asphalt tapers (minimum 40:1) using asphalt patch material (cold mix) with tar paper underneath at all deck joints and bridge ends as required to maintain adequate traffic movement. The asphalt tapers shall be maintained until traffic is no longer using that lane or the final paving is complete. The temporary tapers shall be as per DBSS 501.

413.32.03.02 - Not Used.

413.32.04 Removal of Concrete for Partial and Full Depth Repairs

The Contractor, in the presence of the Quality Manager and Ministry Representative, shall carry out a detailed visual inspection of the deck surface for patch concrete. A chain drag method shall be used to examine the entire deck surface for evidence of additional delaminated and deteriorated concrete. These areas of unsound concrete and patch concrete shall be removed to sound concrete or removed full depth when considered necessary by the Quality Manager and Ministry Representative.

Removal and repair of a specific area will not be completed until the Contractor, the Quality Manager and the Ministry Representative have inspected the affected areas requiring repair, and have agreed on the

extent of repair required.

The Contractor shall supply the necessary labour and supplies to mark out the repair areas.

Concrete for repair for partial and full depth repairs shall proceed as follows:

- The Contractor shall keep repair boundaries square or rectangular and avoid abrupt changes in width of a given repair area. The edge of the repair area shall be maintained vertical. No debris shall be permitted to fall from the bridge. Saw cut the perimeter of the repair areas to a minimum depth of 25 mm to prevent feather edges. Do not cut the existing reinforcing steel.

- Remove deteriorated, delaminated and patch concrete to sound concrete. Maximum jackhammer shall be 18 kg. Care shall be taken to avoid damage to the reinforcement and remaining concrete. Jackhammers are to be operated at angles less than 45º as measured from the surface of the deck to the hand tool. The Contractor shall use wedge type chisel bits. Moyles and round point chisels are not permitted. Adhering concrete shall be removed from the reinforcement using hammers no larger than 7 kg.

- Remove all concrete to allow a 25 mm space around all exposed reinforcing bar within partial depth repairs. Expose all corroded reinforcement at edges of partial and full depth repair area

- Full depth repairs will only apply when the soffit concrete is removed. For full depth repair areas, the Contractor shall install all necessary formwork prior to cleaning of reinforcing and concrete surfaces. The finished soffit repair concrete shall be flush with the surrounding soffit concrete. Areas for full depth repair exceeding 1 m^2 shall be patched prior to placement of the overlay. Before placement of the concrete patch, the surface of the adjoining concrete should be saturated with water for a period of no less than 30 minutes and coated with a bonding agent immediately ahead of the fresh concrete.

- Mildly corroded reinforcement shall be blast cleaned to commercial blast finish SSPC-SP6. Severely corroded and damaged reinforcement shall be replaced. Mechanical couplers acceptable to the Designer or laps meeting the CSA S6-00 Code requirements shall be used to attach new bars to the existing bars. Mechanical couplers shall develop at least 125% of the specified yield strength of the bar.

- Prior to the placement of the partial and full depth repair area infill concrete the surfaces of the sub deck within the repair area shall be abrasive blast (minimum 35 MPa/5000 psi) or high-pressure water blast (minimum 35 MPa/5000 psi) cleaned to remove all bruised and fractured concrete and foreign materials such as dirt, dust, laitance, sand, grease, oil, concrete slurry etc. to the satisfaction of the Quality Manager and Ministry Representative.

- The Contractor shall collect and remove all dirt, dust, and blasting sand from the repair areas of the sub deck, by vacuum or similar method, and dispose of the materials off the bridge site. Disposal shall be in accordance with all applicable environmental regulations.

413.32.05 Reinforcing Steel

Reinforcing steel shall be supplied and installed in accordance with DBSS 412. Welding of reinforcing bars will not be permitted.

Severely corroded reinforcing steel, more than 25% section loss as determined by the Quality Manager, in the existing deck shall be removed and a replacement bar spliced using mechanical couplers acceptable to the Designer or lapped with existing reinforcing steel. Mechanical couplers and lap length shall be in accordance with CAN/CSA S6-00

Reinforcing steel damaged by the Contractor's operations shall be replaced.

413.32.06 Removal and Replacement of Existing Deck Joints

413.32.06.01 Expansion joint system - The Contractor is responsible for the demolition and removal of the existing concrete and expansion Joints.

The steel portions of the deck joint shall be supplied and fabricated in accordance with DBSS 422.

The steel portions of the deck joint shall be galvanized and the joint armoring shall be installed 5 mm below the finished concrete nosing surface. Galvanized surfaces damaged by welding shall be touched up with two coats of a zinc rich coating approved by the Quality Manager..

Each joint seal shall be supplied in a single length, without splices. Before the joint seal is installed, the joint shall be thoroughly cleaned with a wire brush and all moisture removed from the joint. The seal shall be installed in accordance with the Manufacturer's recommendations.

413.32.06.02 Preparation of Filled Joints - The Contractor shall cut a transverse line, directly above the filled joint location, during casting of the deck overlay concrete using an edge cutter attached to a bull float. The line shall be neat and straight.

The Contractor shall further saw cut the filled joint along the transverse line within 48 hours of casting the deck. The saw cut will be achieved using a 4mm wide diamond blade to a depth of 30 mm below the top of the deck.

The Contractor shall fill the saw-cut with Sikaflex 1C SL, a one-component, self-leveling, polyurethane sealant or an alternate permanent filler, impervious to water and resistant to ultra violet radiation, acceptable to the Designer and Ministry Representative. Sealants shall be installed in accordance with the manufacturers' recommendations. The saw cut shall be clean and dry before placing the sealer.

413.32.07 Preparation of the Sub Deck

The Contractor shall high pressure (minimum 110 MPa/15000 psi @ 68 liters/minute using a rotating head) water blast the surface of the entire concrete subdeck including both parapet or curb faces 50 mm up from original deck elevation and abutment ballast walls, no earlier than 2 days before placing the overlay concrete.

The water blasting equipment shall effectively remove laitance, loose materials, bruised concrete, dust, slurry, oil, or other contaminants (as applicable) that are detrimental to the concrete overlay bond and shall leave behind a sound concrete surface, thoroughly cleaned and roughened to partially expose the coarse aggregate. The Contractor shall provide full containment and disposal of all debris, contaminants, etc., and shall comply with all applicable environmental and Workers Compensation Act, Occupational Health and Safety Regulation BC, requirements. The Contractor shall provide adequate guards to contain over spray and flying debris during the scarification, water blasting, deck washing and chipping process. The Contractor shall take all means necessary to reduce and contain the amount of dust produced throughout the entire project.

For the period between water blasting and placement of overlay concrete the sub-deck shall be protected from contaminants.

Water blasting equipment shall always travel downhill, keeping all waste material ahead of the equipment and preventing the waste material from rehydrating on the previously cleaned surface.

413.32.08 Concrete Overlay Placement

All concrete works will be governed by the requirements of DBSS 211 "Portland Cement Concrete" , DBSS 933 "Admixtures for Portland Cement Concrete" and DBSS 413.31 except as otherwise specified in this Section.

The Designer shall be responsible for the design of all concrete mixes. The Quality Manager shall be responsible for quality control and quality assurance of each component of the concreting operation, including aggregate and component quality, batching, mixing, transporting, placing, consolidating, finishing, curing and testing.

The Contractor shall submit a detailed plan of equipment and manpower to be used for placement and curing of the concrete overlay a minimum of 14 days prior to the scheduled placement dates for the review of the Quality Manager and Ministry Representative. The plan shall also contain details such as: water source, the distribution system, the collection and run-off control system, person(s) responsible for quality control, and staff to maintain the system. All review concerns raised by the Quality Manager and the Ministry Representative shall be addressed to the satisfaction of the Ministry Representative.

The Contractor will undertake a test placement of the proposed deck mix design as indicated by the requirements of DBSS 211.03.05.

413.32.08.01 Materials & Equipment - Acceptable types of overlay are High Density concrete and Silica fume modified concrete**.** For High Density concrete overlays, acceptable finishing machines are Bidwell model OF400, Bidwell model OF500 or alternate (low slump pan-type overlay finishers), acceptable to the

Designer and Ministry Representative. Concrete shall be produced on site using volumetric concrete mobile mixers. All equipment shall be provided in good working condition.

For structures over 50 m long and structures over 20 m long with a grade greater than 3%, two finishing machines shall be used in tandem. The speed of the trailing machine shall be such that its motion is continuous. Any open texture remaining after two finishing machine passes shall be repaired by hand.

For Silica fume modified concrete overlays, acceptable finishing machines are Bidwell model 4800, Gomaco model CA450 or alternate acceptable to the Designer and the Ministry Representative. Concrete shall be produced by either a qualified concrete batch plant or at the bridge site using pre-bagging concrete mix and mixer trucks. All equipment shall be provided in good working condition.

Finishing machines shall be capable of forward and reverse motion under positive control and with provision for raising screeds to clear previously screeded surfaces while traveling in reverse.

The Contractor shall provide steel rail mounted mechanical deck concrete finishing equipment of adequate size and design to permit the complete placement and finishing of a single lane section from abutment to abutment with the prescribed surface finish, without forming any cold joints.

The Contractor is responsible for properly setting the screed rails to ensure longitudinal and transverse drainage from the deck without ponded areas or "bird baths".

Sufficient screed guide rails will be set out for the full length of the anticipated section to be cast plus 6 m length at each end for run in and run out of the deck machine, adjusted for height and accepted by the Quality Manager and Ministry Representative prior to the pour as follows:

i) Screed guide rails, upon which the finished machine will be placed outside the area to be concreted, will be horizontally and vertically stable. Rails shall be installed with fully adjustable supports at 350 mm maximum spacing

ii) The finishing machine and guide rails will be adjusted so that the height of the screed above the existing subdeck at each point meets the Quality Manager's and Ministry Representative's requirements. To confirm the adjustment of the machine and guide rails, the screed will be "dry-run" and clearance measurements taken at 3 m intervals and provided to the Quality Manager and, when requested, the Ministry Representative for acceptance. The minimum overlay thickness shall be 50 mm. Resetting of the machine and/or guide rails will be done as necessary to obtain an acceptable "dry-run". Adjustments to the machine or the rails will not be permitted after an acceptable "dry-run" is complete.

Work bridges shall be provided to facilitate bull floating, edge finishing with hand tools, correction of defects in machine finished concrete and tining and fog curing.

Rigid forms made of wood or steel shall be installed at longitudinal bulkheads in a manner that will maintain the desired shape during concrete placement, consolidation and finishing. The longitudinal bulkhead will be installed at the centerline or lane lines of the bridge deck. The top of the longitudinal bulkhead will be set at the finished deck elevation. Any honey combing or poorly consolidated concrete after stripping the longitudinal bulkhead will be cut back into the new overlay a minimum of 150 mm.

The overlay shall be placed in strips with longitudinal joints near lane markings. Each strip shall be placed continuously between joints; adjacent strips shall not be cast for 24 hours.

413.32.08.02 Bonding Agent - The Contractor shall apply a cementitious bonding slurry, pre-approved by the Designer, to the prepared sub-deck and repair concrete. The prepared concrete sub deck shall be in a saturated surface dry (SSD) state prior to slurry placement. The Contractor shall supply the necessary equipment such as water hoses and vacuum or oil free compressed air.

The cementitious bonding slurry is to consist of type GU portland cement mixed with a water/cement ratio not exceeding 0.38 by mass. The slurry must not exceed 3 mm in depth and be scrubbed into the substrate surface with coarse brooms; in particular no pooling will be permitted. The bonding slurry is to be applied immediately ahead of the overlay concrete placement and in a timely fashion so as not allow drying or setting up of the slurry prior to the placement of overlay concrete. Any areas to which the slurry has been applied, that in the opinion of the Quality Manager and Ministry Representative exhibit drying or setting shall be cleaned of the bonding slurry and re-coated prior to the placement of deck overlay concrete.

The vertical faces at the edge of the scarified sub-deck surface; at construction joints; and at concrete/deck-

joint interfaces for old bridge decks and the vertical faces at sub-deck and curb or parapet interface for new decks shall be coated with an epoxy bonding agent conforming to ASTM C881, Type V, Grade 2. The Class of bonding agent shall be appropriate for the temperature at the time of application. The bonding agent shall be applied in accordance with the manufacturer's instructions.

413.32.08.03 Interim Partial Strength - Overlay shall attain a minimum strength of at least 30 MPa before removal of the burlap and opening to traffic. All other curing requirements as specified in DBSS 413.31 shall be adhered to. The interim strength tests shall be determined using field cured test cylinders cured in accordance with Clause 7.3.4 of CSA A23.2-3C.

SECTION 415

MANUFACTURE AND ERECTION OF PRECAST AND PRESTRESSED CONCRETE MEMBERS

DESCRIPTION

415.01 Scope – This Section covers the manufacture and erection of precast and prestressed concrete members

415.02 General

415.02.01 The manufacture and erection of precast and prestressed concrete members shall conform to the requirements of the Contract and the Design.

415.02.02 Precast concrete elements shall be manufactured in plants certified to the current CSA-A23.4 in the appropriate category by a certification organization accredited by the Standards Council of Canada in the subject area of building products and structures. Certification shall be in effect prior to the beginning of Work , and maintained throughout the period of manufacture.

415.02.03 Materials and construction shall conform to the current CSA- A23.4 and the current PCI "Manual for Quality Control for Plants and Production of Precast and Prestressed Concrete Products ".

415.03 Prefabrication Meeting – The Quality Manager will convene a prefabrication meeting with the Contractor, fabricator and the Ministry Representative to review procedures on quality control, quality assurance, plant certification, concrete source and mix designs, use of Hydration Stabilization Admixtures (if applicable), mill certificates, alternate details and procedures, updated schedule for prefabrication and to confirm the Contract and Design requirements.

415.04 Working Drawings - Working drawings shall consist of the following:

- Shop drawings,
- Transportation details, and
- Erection drawings

Transportation details and erection drawings shall be prepared and sealed by a professional engineer registered with APEGBC.

Shop drawings shall be prepared and sealed by a professional engineer registered with the Association of Professional Engineers and Geoscientists of British Columbia (APEGBC) when the Contractor is responsible for the design of items that are detailed on the shop drawings.

Working drawings shall be in the same system of units as the Design.

Working drawings shall be on D size sheets approximately 560 mm (22") by 865 mm (34"), (one drawing per sheet). Lettering for notes and dimensions shall be at least 2.5 mm and 4 mm for headings. Drawings shall be legible when half-sized or microfilmed.

415.04.01 Shop Drawings –Shop drawings shall show all information and details needed for the fabrication of the members including, but not limited to, such items as member shapes and dimensions, mark numbers and general arrangement of member locations, mass, prestressed and non-prestressed reinforcement, embedments, openings, block outs, chamfers, recesses, finishes, concrete mix design, stressing details, special tolerances, special handling instructions, lifting details and lifting locations.

415.04.02 Transportation Details – Transportation details shall include such items as:

- Description of hauling and handling equipment,
- Weight of members,
- Length and height of loads,
- Location and method of member support, and engineering calculations where members are to be supported during transportation further from their ends than outlined in DBSS 415.48,
- Details for handling, storing, and loading of members.

415.04.03 Erection Drawings

Erection drawings shall show in detail the method of erection including, but not limited to, the following:

- Erection procedures
- Procedures for off loading of members upon delivery
- Details for temporary storage and support of members on site prior to erection
- Equipment to be used
- Layout or general arrangement drawing showing the layout of the members, equipment positioning, and access roads
- Crane make, model, and capacity charts, boom length(s) , crane placement, and access for transporting of members to crane(s)
- Radii and loads for crane lifts
- Rigging details
- Mass of members , rigging and special installation equipment
- Details for installation and removal of all falsework, temporary supports, temporary bearings, bracing, guys, dead-men, and lifting devices
- Attachments to the bridge members and bridge structure for temporary support and special launching

equipment
- Detailed description of sequence of operations
- Details for special installation equipment such as a launching truss, launching nose, head frames, spreader beams and rollers
- Details for installation of members onto the permanent bearings
- Traffic control plan for roadway and rail traffic
- Details for protection of existing utilities affected by the erection procedures
- Layout and detail of fall protection and their sequence of installation.

The Contractor shall be responsible for the lateral stability of members and shall design and provide bracing as necessary until completion of the Work.

The erection drawings shall be complete in detail for all anticipated phases and conditions during erection and during the temporary support of members. The Contractor shall submit calculations upon request, to the Designer and Ministry Representative, that demonstrate that allowable stresses are not exceeded in members, falsework, temporary bracing and temporary supports and that member capacities and final geometry will be correct. The calculations shall be sealed by the professional engineer that sealed the erection drawings.

Falsework, temporary supports and temporary bracing shall meet the requirements of CSA Standard S269.1, "Falsework for Construction Purposes" and shall also meet all the requirements for falsework given in the applicable Clauses of the Workers Compensation Act, Occupational Health and Safety Regulations BC.

A professional engineer registered with APEGBC shall be responsible for any field designs and any changes made to the erection procedures. Field designs and changes to the erection procedures must be documented and sealed by the responsible professional engineer and must be available at the Site prior to the affected erection work being carried out.

Immediately before placement of loading on falsework, the Contractor must ensure that the falsework is inspected and a sealed engineering certificate is issued by a professional engineer registered with APEGBC which:

- Indicates the specific areas inspected, and
- Certifies that the falsework has been erected in accordance with the latest approved erection drawings and supplementary instructions.

Prestressed concrete stringers shall not be set on their bearings, nor shall the bridge deck be cast, until 30 days after the casting of the stringers. If it is necessary to erect the stringers earlier, they shall be placed on temporary supports. Prestressed concrete twin cell box stringers up to a maximum length of 20 metres may be shipped and erected after 10 days if the specified 28 day compressive strength has been attained.

415.04.04 Submittals - The Contractor shall submit to the Designer copies of all working drawings. Copies shall also be provided to the Ministry Representative. These drawings shall be submitted at least fourteen days prior to the fabrication of the work. The Designer will review the drawings for general compliance with the Design and the Contract.

If modifications to the drawings are required, the Designer will return one set of drawings, marked up, to the Contractor. The Contractor shall re-submit copies of revised drawings to the Designer for review. Copies shall also be provided to the Ministry Representative.

If no exceptions are taken to the drawings, the Designer will return one set of reviewed drawings to the Contractor. The Contractor shall forward two sets of the reviewed drawings to the Ministry Representative.

Shop drawings will not be reviewed without the transportation and erection drawings applicable to the members in question.

Any work done or materials ordered prior to the review of working drawings shall be at the Contractor's risk. Erection will not be allowed to proceed without the Designer's review of the method proposed.

Review of working drawings shall not relieve the Contractor of any responsibility for dimensions or detail or for carrying out the work in full accordance with the Contract and the Design.

415.05 Alternative Details - All details shall, in general, conform to those shown on the Design. Subject to the Designer's consent and with notification to the Ministry Representative, alternative details may be substituted to facilitate the Contractor's work methods. Such alternative details shall comply in all respects with this specification.

415.06 Quality Control - The Contractor shall implement a quality control program to meet the Contract requirements.

415.07 Quality Assurance and Audit - The Quality Manager will implement a quality assurance program by auditing the Contractor's quality control program and by inspection at its discretion. The Ministry will implement a quality audit program by auditing the quality control and quality assurance programs implemented by the Contractor and by inspection and testing at its discretion.

The Contractor shall notify the Quality Manager and Ministry Representative at least 14 days before fabrication is to commence. The Contractor shall allow the Quality Manager and Ministry Representative access to all parts of work, and shall supply such information and assistance as is required. When the Quality Manager or Ministry Representative requests, the Contractor shall provide samples of any materials. Inspection by the Quality Manager or audit by the Ministry Representative shall not relieve the Contractor from obligation to perform the work in accordance with the Contract and the Design.

Concrete tests shall be in accordance with the current CSA A23.2. The Contractor shall make available at the plant a sheltered, heated space for the casting and storage of test cylinders, with curing tanks and a concrete cylinder testing machine.

The Quality Manager or Ministry Representative may reject any items which, in its opinion, do not comply with the requirements of this specification.

Each unit shall be certified as acceptable by the Quality Manager before it is shipped from the shop. This certification shall not relieve the Contractor of responsibility for subsequent damage or for defects which become apparent before the Work is finally accepted.

MATERIALS

415.11 Materials Supplied by the Contractor

415.11.01 Cement - Portland cement shall conform to the requirements of CAN/CSA-A3001, Cementitious material for use in concrete.

415.11.02 Aggregates - Fine and Coarse Aggregates shall conform to the current CSA A23.1. Coarse aggregates shall not exceed 20 mm nominal size.

415.11.03 Water - Water shall conform to the current CSA A23.1. The mortars and tests shall be made in accordance with the current CSA A23.2.

415.11.04 Reinforcing Steel - Reinforcing steel shall be in accordance with the current CAN/CSA G30.18, G30.3 and G30.5.

All reinforcing shall be of the lengths called for on the Design and all bends shall be made in accordance with DBSS 412.

All bending schedules shall be furnished by the Contractor for the Designer's review and, if requested a copy shall be provided to the Ministry Representative.Any material ordered before such review shall be at the Contractor's risk.

415.11.04 Epoxy-Coated Reinforcing Steel - All reinforcing steel labelled "E" on the drawings shall be epoxy-coated to the requirements of ASTM A 775 M and D 3963M.

415.11.05 Prestressing Steel – Pre-tensioning steel shall consist of seven-wire, uncoated, low relaxation strands and shall comply with the requirements of the current ASTM A 416M.

All strands shall be obtained from one mill, and preferably strands from one heat shall be used. Each reel of strand shall be identified by a secure tag showing the size of strand, CSA or ASTM Designation number, heat number, and name or mark of the manufacturer.

The Contractor shall supply to the Quality Manager and, if requested, to the Ministry Representative, for each heat number, one set of the standard mill test reports. Strands shall be protected at all times to prevent corrosion and shall be dry and clean (free from scale, rust, oil, soap, grease, and other deleterious materials) immediately before the concrete is placed. No strand which has previously been tensioned or gripped by strand chucks shall be incorporated into the members.

415.11.06 Admixtures – Chemical admixtures shall conform to the current ASTM C-494. Air-entraining admixtures shall conform to the current ASTM C-260.

Water reducing admixtures shall be of a lignosulfonic acid or salt, or hydroxylated carboxylic acid or salt.

Type B, Retarding or Type D, Water-Reducing and Retarding (Hydration Stabilizing) Admixtures conforming to the requirements of ASTM C494 shall not be incorporated into the mix design and/or added to the concrete without the acceptance of the Quality Manager and Ministry Representative. When accepted, the guidelines given in DBSS 415.37, DBSS 415.40.01 and DBSS 415.42.01 shall apply.

415.12 Storage of Materials - All materials shall be stored in accordance with the requirements of the current CSA A23.1. Cement shipments shall be used in the order in which they are received.

415.13 Premixed Aggregates - Pre-mixed aggregates in which the fine and coarse aggregates are combined in definite proportions will be permitted provided that the aggregates are delivered in batches directly to the hopper of the concrete mixer. Pre-mixed aggregates shall be transported and delivered in batches, each containing the correct quantity for one batch of concrete. At no time after the fine and coarse aggregates are combined shall any batch

come in contact with or intermingle with any other batch.

415.14 Forms - All exterior forms shall be of steel and shall be mortar-tight and of sufficient rigidity to prevent distortion due to incidental loadings during construction. Forms for skewed ends may be of rigid steel-reinforced plywood.

Wood forms will be permitted for the side walls of ballast wall pieces and for side walls of the deck panel pieces. Wood formwork shall be designed supplied and installed in accordance with CAN/CSA-S269.3-M. Wood formwork shall render a true smooth surface free from fins and projections. Only new plywood shall be permitted for plywood formwork.

Forms shall produce 20 mm chamfers and fillets as shown on the Design.

415.15 Ministry's Office - The Contractor shall provide at the point of manufacture a suitable lock-up office for the sole use of the Ministry Representative.

The office shall be weatherproof and reasonably soundproof and provided with ample window area and ventilation. Location of the office shall be acceptable to the Ministry Representative. The office shall measure not less than 3 m x 4 m. In the event that the office is to be used concurrently with other inspection agencies, the minimum area required for Ministry's use shall be 3 m x 4 m. The Contractor shall install satisfactory heat and lights and provide a telephone and a draughting table 1 m x 2.5 m in size with a plywood top, desk 0.75 m x 1 m, two chairs, drafting stool and one square metre of shelves.

Long distance calls made by Ministry personnel will be to the Ministry's account.

The Contractor shall, during the life of the work, regularly clean and properly maintain, heat and light to the office.

The office and contents shall be for the use of the Ministry Representative for the duration of the Work and may, if necessary, be used concurrently with other inspection agencies.

CONSTRUCTION

415.31 Reinforcing Steel - All reinforcing bars shall be accurately placed in the positions shown on the Design and firmly held during placing and setting of the concrete. Tolerances shall be as specified by the current CSA A23.4.

When placed in the work the reinforcing steel shall be free from dirt, rust, loose scale, paint, oil, or other foreign materials. Bars shall be tied at all intersections except that where spacing is less than 300 mm, alternate intersections shall be tied. Form clearance shall be maintained by stays, ties, hangers or other approved supports. Before placing concrete, the placing and securing of reinforcing in the forms shall be inspected by the Quality Manager.

All reinforcing shall be placed in the full lengths called for on the Design. Splicing of bars will not be permitted unless acceptable to the Designer. Splices shall be staggered. Unless otherwise shown or called for on the Design, splices shall be made by lapping in accordance with the requirements of DBSS 412. The bars shall be placed so as to maintain the minimum specified clearances between bars and the concrete surface. Welding of reinforcing steel shall be done only if called for on the Design or accepted by the Designer and Ministry Representative.

Tack welding of reinforcing steel, if permitted, shall be in accordance with the current CSA W186-M.

415.32 Stressing - The dials of gauges shall be at least 200 mm in diameter. Gauges shall be calibrated with the hoses and jacks with which they are to be used, by an approved testing agency. The calibration must have been performed within the previous 6 months of use or 12 months maximum. Before stressing commences, certified calibration curves shall be furnished to the Quality Manager and Ministry Representative if requested.

If during the work any gauges give erratic results, the tension system or tension device shall be recalibrated.

The Contractor shall schedule the work so that placement of concrete follows tensioning of strands within 24 hours. In cases of unavoidable delay, a maximum period of 80 hours will be allowed, provided that the strands do not show unacceptable corrosion (see current PCI “Manual for Quality Control for Plants and Production of Precast and Prestressed Concrete Products”).

Before tensioning, an initial force shall be applied to each strand to equalize stresses and eliminate slack. This force shall not exceed 22 kN. After the initial tension forces have been applied, each strand shall be marked at the outer end of each chuck as a reference line from which the elongation of the strand can be measured.

The method of tensioning shall be such that the specified tension is maintained in the strands until the forces in the strands are transferred to the concrete member(s). The maximum jacking load in each strand shall be 80% of the ultimate strength of the strand.

A tensioning force shall be determined both by reading the jacking load on the calibrated gauge and by measuring the elongation of the strand. When the difference between the

two methods is greater than 5%, the cause of the discrepancy shall be ascertained and corrected.

Elongations of strands shall be calculated and shown on the working drawings. Calculations for elongations shall include appropriate allowances for possible slippage or relaxation of the anchorages, friction at hold down points, temperature difference, and strand relaxation.

415.33 Forms - The design of the forms shall take into account the effect of vibration of concrete as it is placed.

All exterior forms shall be set and maintained true to the designated lines until the concrete has hardened. When forms are unsatisfactory in any way, either before or during the placing of concrete, the Quality Manager or Ministry Representative may order the work stopped until the defects have been corrected.

The shape, strength, rigidity, mortar-tightness and surface smoothness of re-used forms shall be maintained at all times. Any warped or bulged formwork must be realigned before being re-used. Forms which are unsatisfactory in any respect shall not be reused.

Void forms shall be of an approved product and shall be anchored firmly by a method acceptable to the Quality Manager and Ministry Representative, to prevent displacement during the placing of the concrete. Any member with a void out of position in excess of the specified tolerance will be rejected.

Exterior forms shall be treated in moderation with form oil before placing the concrete. Any material which will adhere to or discolour the concrete shall not be used.

Side forms and any attachments which would obstruct strain of the members shall be removed before transfer of stress.

415.34 Admixtures - The Contractor shall add sufficient air-entraining agent as required by the Design as determined by an air-meter used in accordance with the current CSA A23.2.

The use of other admixtures, such as water-reducing agents or superplasticizer, shall be subject to the acceptance of the Designer and Ministry Representative.

Type B, Retarding or Type D, Water-Reducing and Retarding (Hydration Stabilizing) Admixtures shall not be incorporated into the mix design and/or added to the concrete without the acceptance of the Ministry Representative. When accepted by the Ministry Representative, the guidelines given in DBSS 415.37, DBSS 415.40.01 and DBSS 415.42.01 shall apply.

415.35 Measurement of Materials - Measurement of materials shall conform to the current CSA A23.1. The Contractor shall produce a certificate or report of inspection from a weights and measures approved, independent testing authority, and which is not more than 6 months old. The certificate or report will be accepted as proof of accuracy for the batching equipment. Cement in standard packages need not be weighed.

The surface moisture shall be determined in accordance with the current CSA A23.2 and the batch weights of aggregates and water adjusted accordingly.

415.36 Mix Requirements - Minimum compressive strength at time of transfer of stress to the concrete and at 28 days as determined by the current CSA A23.2 shall be as called for on the Design. In any case, the force in the prestressing tendons shall not be transferred to the concrete until the compressive strength of the concrete is at least 25 MPa for pretensioned components and 20 MPa for post-tensioned components.

Slump measurement shall be determined by the current CSA A23.2. The mix design shall specify the slump and the tolerances in the measured slump shall be ±20 mm for superplasticized concrete. Slump shall not exceed 200 mm for superplasticized concrete and shall be measured within 5 minutes after discharge from the mixer.

In order to minimize shrinkage and creep the cement/aggregate ratio shall be kept as low as is consistent with strength requirements.

415.37 Batching and Mixing - The batching plant and mixing of concrete shall conform to the current CSA A23.1.

Mixing shall begin within 15 minutes after the cement has been added either to the water or aggregate.

Within 30 minutes after the introduction of the mixing water to the cement and aggregate, or the cement to the aggregate, the concrete shall be placed in its final position in the forms. In hot weather or under other conditions contributing to quick stiffening of the concrete, the Contractor shall modify work procedures as necessary to reduce the time of placement

Time of placement extensions will be considered on a case by case basis and may require the use of Hydration Stabilizing Admixtures (HSA's) conforming to the requirements of ASTM C494 Type B, Retarding or Type D, Water-Reducing and Retarding Admixtures. If accepted by the Ministry Representative, guidelines for the use of HSA's shall be as follows:

- HSA modified concrete when used shall be fully discharged and placed in it's final position within 90 minutes after the introduction of water to the cement and aggregate.

- All concrete in a given precast element shall be placed in it's final position before 90 minutes.

When HSA's are used, these time extensions are subject to preconstruction trials being conducted by the Contractor, to establish the appropriate HSA dosage to provide suitable extended slump life of concrete, without increasing the water/cementitious ratio of the concrete above that which would be required if HSA's were not used. The use of HSA's shall in no instance modify the maximum concrete temperature required at time of placement. A one time only addition of HSA will be allowed. This will be during initial batching of the concrete or immediately at completion of batching as recommended by the admixture manufacturer. Addition of HSA at any other time will be cause for rejection of the concrete.

Concrete mixing and transporting equipment shall be capable of placing concrete in the forms at the rate of 20 cubic metres per hour. Batch delivery shall be uniform and at a maximum interval of 10 minutes.

The maximum size of each batch shall not exceed the maximum rated capacity of the mixer as stated by the manufacturer of the mixer.

Concrete when placed in the form shall have a temperature not less than 10°C nor greater than 25°C.

415.38 Handling and Placing Concrete - In preparation for the placing of concrete, all construction debris and extraneous matter shall be removed from the interior of forms. Struts, stays and braces inside the forms shall be kept to a minimum and entirely removed when the concrete placing has reached an elevation rendering their service unnecessary.

Concrete shall be handled and placed in accordance with the requirements of the current CSA A23.1 and shall be placed in the shortest possible time after mixing is completed. When rectangular void forms are used, concrete under voids shall be placed and compacted before void forms are in position.

Concrete may be deposited a maximum of 2 m without the use of pipes, provided, when required, suitable measures are taken to prevent segregation and premature coating of upper reinforcing steel. When pipes are used they shall, as far as practicable, be kept full of concrete during placing and their lower ends shall be kept buried in the newly placed concrete.

After initial set of the concrete, the forms shall not be jarred and no stress shall be placed on reinforcing bars which project.

415.39 Vibration - Concrete, during and immediately after depositing, shall be thoroughly compacted. The compaction shall be accomplished by mechanical vibration subject to the following provisions:

a) Vibration shall be internal for all accessible parts and shall be external for inaccessible parts.

b) Vibrators shall be of a type and design acceptable to the Designer and Ministry Representative. They shall be capable of transmitting vibration to the concrete at frequencies recommended in the current CSA A23.1.

c) The intensity of vibration shall be such as to visibly affect a mass of concrete of 50 mm slump over a radius of at least 500 mm.

d) The Contractor shall provide a sufficient number of vibrators to properly compact each batch immediately after it is placed in the forms.

e) Vibrators shall be manipulated so as to thoroughly work the concrete around the reinforcing and embedded fixtures and into the corners and angles of the forms.

Vibration shall be applied at the point of deposit and in the area of freshly deposited concrete. The vibrators shall be inserted into and withdrawn from the concrete slowly. The vibration shall be of sufficient duration and intensity to thoroughly compact the concrete, but shall not be continued so as to cause segregation. Vibration shall not be continued at any one point to the extent that localized areas of grout are formed.

Application of vibrators shall be at points uniformly spaced and not farther apart than twice the radius over which the vibration is visibly effective.

f) Vibration shall not be applied directly or through the reinforcement to sections or layers of concrete which have hardened to the degree that the concrete ceases to be plastic under vibration. It shall not be used to make concrete flow in the forms over distances so great as to cause segregation, and vibrators shall not be used to transport concrete in the forms.

g) Vibration shall be supplemented by such spading as is necessary to ensure a smooth surface and dense concrete along form surfaces and in corners and locations impossible to reach with the vibrators.

h) Vibrator heads shall be smaller than the minimum clear distance between prestressing strands.

415.40 Test Cylinders - The Contractor shall make available, as may be required by the Quality Manager or Ministry Representative, samples of the concrete to be used in casting the members.

Further, the Contractor shall take samples and make test cylinders in the presence of the Quality Manager and Ministry Representative and in accordance with the current CSA A23.2-1C, -3C and -9C. At least 8 cylinders per casting bed shall be made for testing.

415.40.01 Release Strength - Two cylinders from each end of the casting bed shall be cured in the same manner as the members and tested by the Contractor under the discretionary observation of the Quality Manager or Ministry Representative (QM or MR) at the Contractor's plant to assess the strength of the concrete at the time of transfer of force to the concrete. Only one cylinder from each end need be tested if the average strength of the two cylinders is equal to or greater than the required release strength at transfer and no test cylinder is more than 1.4 MPa below the release strength. No transfer of force shall occur without the acceptance of the Quality Manager.

If all release cylinders are tested and the release strength has not been attained, further testing with an impact hammer shall be allowed, provided a written procedure acceptable to the Designer and Ministry Representative is provided by the Contractor. The procedure shall correlate readings from areas of a prestressed girder where the release strength has been determined by concrete test cylinders with similar areas on the girder in question. If this situation occurs, extra pairs of release cylinders shall be made on future casts until confidence in determining the release strength is re-established.

When HSA's are incorporated into the mix design, pre-construction tests shall be conducted to determine the rate of early strength development for the selected concrete mix design. Tests shall be repeated when there is a change in the HSA dosage rate.

415.40.02 28 Day Strength - Two cylinders from each end of the casting bed shall be identified as "acceptance cylinders" and shall be cured in the Contractor's curing tanks. These cylinders shall be tested by the Contractor under the discretionary observation of the Quality Manager or Ministry Representative to determine the twenty-eight day strengths. The strength of the concrete in the member shall be taken as the average of the compressive strength of the "acceptance cylinders", however, no individual cylinder shall be more than 3.5 MPa below the specified strength and the average strength shall equal or exceed the 28 day strength.

415.40.03 Additional Cylinders - For special cases, extra cylinders shall be made and tested if required by the Quality Manager or Ministry Representative. In the case of ready mix, and plant mixed concrete with no established records, extra cylinders, in addition to the release strength or 28 day strength cylinders, shall be made from each load.

415.41 Surface Finish - Throughout the work, special precautions shall be employed in formwork and concrete placing to produce a smooth and dense concrete. Exposed surfaces shall be as free as possible from form marks, fins, honey-combing and pock marking. Only a minimum amount of pointing will be allowed. Poor surface finish upon the removal of the forms shall be sufficient cause for the rejection of any member. All exposed edges shall be chamfered unless otherwise noted on the Design.

Roadway and surfaces to be bonded to shall be rough screeded by wood floating following initial strike off. Surfaces to be bonded to shall be sandblasted or exposed with chemical retarder and water blasted to remove all laitance and foreign matter and to roughen the surface. The amplitude of the surface roughness shall be equal to one third of the nominal size of the concrete coarse aggregate or 6 mm, whichever is less. Loose sand and debris shall be washed or blown off. Water, cement or proprietary topping material shall not be added to the surface during finishing.

Walkway surfaces shall be given a coarse transverse broomed finish following initial screeding and floating. Roadway surfaces not requiring any wearing surface shall be given a tined surface in accordance with DBSS 413.31.02.05. Roadways receiving a membrane shall be finished in accordance with DBSS 419.33.

The outer surfaces of the outside stringers and any other surfaces indicated on the Design shall be given a Class 2 finish in accordance with DBSS 211.17.

415.42 Curing - Members may be cured by steam or heat as outlined herein.

415.42.01 Steam Curing: The concrete shall be allowed to attain its initial set before application of steam (generally 2 to 4 hours after the final placement of concrete or as justified by previous experience and acceptable to the Quality Manager and Ministry Representative.

If HSA's are used in the mix design, time to attain initial set after concrete has been placed may exceed that required for conventional concrete. The concrete shall have attained initial set before the application of steam.

After the placement of concrete, sufficient steam shall be applied during initial curing to hold the air surrounding the member at a temperature not less than 13°C.

Steam curing shall be done within a suitable enclosure to contain the steam and minimize moisture and heat losses. The enclosure shall allow free circulation of steam around the sides and tops of members and shall not be in contact with the member.

Steam at the outlets shall be at 100% relative humidity and low pressure. Application of the steam shall not be directly on the concrete or test cylinders and shall be from outlets distributed uniformly along the whole length of the member.

During application of the steam the ambient air temperature shall increase at a rate not to exceed 22°C per hour until an optimum temperature of 55°C is reached. In no case shall the ambient curing temperature exceed 80°C. The maximum temperature shall be held until the concrete has reached the transfer strength. In discontinuing the steam, the ambient air temperature shall not decrease at a rate greater than 33°C per hour until a temperature has been reached about 11°C above the temperature of the air to which the concrete will be exposed. The forms shall not be stripped until the completion of this cooling off period.

Provision shall be made for checking the ambient air temperature either by placing minimum-maximum thermometers under the enclosure or periodically checking with hand thermometers.

415.42.02 Curing With Radiant Heat and Moisture - Curing with radiant heat and moisture shall be done by heating the casting area in combination with the continuous application of moisture.

Heat shall be applied by means of pipes circulating steam, hot oil or hot water or by electric heating elements. Pipes shall not be in contact with concrete, form surfaces or test cylinders.

Moisture shall be applied by a continuous sprinkling to a cover of burlap or cotton matting or any other approved method for keeping the members continuously moist.

An auxiliary cover in addition to the moisture retaining cover shall be used above the entire bed to retain the heat. This cover shall be supported at a sufficient distance above the member to allow circulation of the heat.

The cycle of application of heat, temperature limitations, reduction of temperatures and use of recording thermometers shall be similar to the cycle specified for steam curing.

415.43 Draped Strands - Where called for on the Design, strands shall be draped in accordance with the current CSA A23.4.

415.44 Transfer of Force to Concrete Members - The transfer of the forces in the strands to the concrete member(s) shall be in accordance with the current CSA A23.4 (section on Detensioning).

415.45 Protective Coating for Ends of Strands - After completion of concrete curing the ends of tensile strands shall be ground flush with the concrete surface. Where the ends of the members are to be embedded in concrete, the ends of the strands shall be painted with two coats of an organic zinc rich paint acceptable to the Designer and Ministry Representative. Where the ends of the members are not to be embedded in concrete, the ends of the members shall be covered with thixotropic epoxy to provide at least 3 mm cover for a band width of 50 mm on each side of all strands. This material is to be applied in accordance with the manufacturer's recommendations.

415.46 Tolerances - For bridge stringers, the tolerances specified in the PCI “Manual for Quality Control for Plants and Production of Precast and Prestressed Concrete Products 1999" shall be modified so that the length of each member, measured parallel to the longitudinal axis, to extreme corners, at 20°C, does not exceed the specified length by more than 10 mm / 10 m (25 mm maximum).

The Contractor shall employ an adequate method of checking tolerances both before and after placement of concrete. Review of the method of checking tolerances by the Quality Manager and Ministry Representative will not limit the Contractor's responsibility for the accuracy of the members.

415.47 Repairs to Members - Defective areas should be repaired immediately following removal of forms and before detensioning of strands. Holes, honeycomb spots, etc., shall be treated as for Class 2 finish in accordance with DBSS 211.17. The method of repair for cracks, broken corners and edges, bulges and other defects shall be submitted to the Designer and Ministry Representative for review.

415.48 Handling, Storing and Loading of Members - Stringers shall at all times be kept in their normal upright position during handling, storage and transportation. They shall be picked up by means of vertical forces or forces inclined towards the opposite ends of the members by no more than 30º from the vertical, applied to lifting devices near the ends of the stringers. I-beams shall be adequately stiffened during lifting and transportation.

Stored stringers shall rest on unyielding level supports near the ends of the stringers.

When loaded for transportation, members shall be supported so that they will not be overstressed. In general, beams with straight strands may be supported up to one-tenth of their

lengths from each end, whereas beams with draped strands should be supported at their bearings. If the Contractor believes that it is possible to safely support members further from their ends, the Contractor shall submit to the Designer and Ministry Representative, sketch(es), sealed by a registered professional engineer, showing the proposed support arrangement. Adequacy of support shall be based on the following assumptions:

- l00% impact on cantilevering ends, when checking stresses at supports,
- 50% negative impact on the whole member, when checking stress near midspan,
- 50% of prestress losses at time of shipping (unless circumstances dictate otherwise),
- allowable tensile stress = $0.5\sqrt{f_c}$
- allowable compressive stress = $0.6f_c$
- where f_c = estimated strength of concrete at time of shipping.

Such sketch(es) shall be submitted at the time of submittal of the working drawings (see DBSS 415.04). Members transported on supports further from their ends than outlined above, without the submittal of sketch(es) acceptable to the Designer and Ministry Representative, will be rejected.

Piles shall be lifted in the horizontal position at the pick-up positions shown on the drawings and at no other points. When stacking for storage or for delivery, piles shall be supported at the pick up points. They shall be handled carefully, avoiding dropping and jarring and avoiding surface abrasions or other damage.

Any member damaged so as to be, in the opinion of the Designer or Ministry Representative, unfit for the use for which it is intended shall be replaced.

415.49 Placement of Concrete in Keyways - Keyways between box stringers shall be clean and shall be saturated surface dry surfaces, before being filled with concrete.

Concrete in keyways shall attain a strength of at least 15 MPa before light traffic such as one-tonne trucks are allowed on the bridge, and at least 25 MPa before heavy loads such as concrete trucks are allowed on the bridge.

415.50 Post-Tensioning - All post-tensioning shall be done in accordance with the DBSS 416 - "Manufacture and Installation of Post-Tensioned Concrete Members".

QUALITY CONTROL PROGRAM

415.61 Quality Control Program – These Subsections describe the general Quality Control Program required for any Fabricator undertaking the fabrication of permanent precast concrete structural components. The Quality Control Program shall be part of the overall Quality Management Plan for the project.

The term "Quality Control" defines those activities that the Fabricator performs to conform to the Contract.

The term "Quality Assurance" defines those activities that the Quality Manager performs to ensure conformance to the Contract.

The term "Quality Audit" defines those activities that the Ministry performs to provide confidence that the QC and QA processes and resulting products satisfy the Contract.

These Subsections contain statements of the Quality Objectives and Policies that the Ministry considers essential for successful and economical quality management. They also outline the Procedures and Documentation to implement and confirm that the Objectives are met.

The provisions of the Quality Control Program set forth in these Subsections shall apply to all precast concrete structural components.

These Subsections require the establishment of a Quality Management Team with the responsibility for the successful and timely implementation of all necessary Quality Control and Quality Assurance activities. Some positions shown in the Suggested Organization Chart (see Figure 1) may be held by the same individual. For example, the General Manager may also be the Contract Administrative Manager and the Purchasing Manager. The Plant Superintendent may also be the Receiver and the Structural Supervisor.

415.62 Quality Objectives and Policies

415.62.01 Quality Objectives - The Quality Objectives of the Ministry of Transportation cover all precast concrete structural components produced under the Contract, as follows:

a) Completed products shall conform fully to the governing Codes and Specifications stipulated in the Design and the Contract.

b) The Quality Control Program shall be fully integrated into the ongoing manufacturing activities of the Fabricator.

415.62.02 Quality Policy - It is Ministry policy that the interests of the Contractor and the Ministry coincide when a product meets the Quality requirements of the Contract and is delivered on time.

The Quality Control Program is an essential part of this policy. It is based on the fact that the best way to do any job is to do it right the first time. This applies to all stages of the

design, drafting, procurement, manufacturing and testing of the product.

The Quality Control Program involves the Quality Manager, Ministry Representative and Fabricator personnel. Their activities must be consistent with the Quality Objectives of the Ministry.

A Fabricator's Quality Control Manager shall be appointed with defined responsibilities in resolving quality matters and shall report to a senior management level. At each hold point, the Fabricator's Quality Control Manager shall:

- Document the successful completion of each stage as it progresses through fabrication and erection.
- Identify and report nonconforming components.
- Initiate or recommend disposition of nonconforming components.
- Verify corrections.

Any persons assigned to perform quality control inspections shall be other than those performing or directly supervising the work and they shall not report directly to immediate supervisors responsible for producing the work.

The Fabricator's Quality Control Program is not subordinate to any design, drafting, procurement, manufacturing and testing activities, it is an essential part of them.

415.63 Scope of the Quality Control Programs - This Quality Control Program governs the fabrication of precast concrete structural components for the Work. The Suggested Organization Chart is shown in Figure 1

The Fabricator's General Manager shall be responsible for:

- adhering to this Quality Control Program in all respects.
- ensuring that completed structural components shall conform fully to the Design and fabrication requirements stipulated in the Contract.
- ensuring that all required documentation is produced according to this Quality Control Program.

415.64 Range of Capability - Within the Company, there shall be the necessary knowledge, skill (in-house or on a contract basis) and equipment to do the following work on precast concrete structural components:

- Design of lifting and erection devices not shown on drawings
- Preparation of shop fabrication drawings
- Preparation of bills of material
- Preparation of material requisitions and purchase orders
- Receiving, checking and storing materials for bridges
- Layout, form set-up, reinforcing steel, prestressing and post-tensioning
- Mix designs, batching, mixing, placing and curing of concrete
- Sampling, testing, making and breaking of concrete test cylinders
- Dimensional checking and verification
- Resolution of non-conformances
- Documentation of all stages of work with capability of tracing all major components
- Finishing, patching, storing and shipping
- Erection of structural components

415.65 Drawings and Specifications – The Fabricator's Chief Design Engineer shall be responsible for:

- the design and location of lifthooks and the preparation of stressing sheets for prestressing or post-tensioning; and for the design of the erection procedures and any special erection equipment needed.

The Fabricator's Chief Draftsperson shall be responsible for:

- obtaining the latest revision of the Design and specifications for the work.
- submitting shop drawings and erection drawings to the Designer for acceptance before commencement of the work.
- preparing material requisitions containing a full description of the material sizes, material specifications, and certifications required for conformance to the contract.
- delivering the material requisitions to the Fabricator's Purchasing Manager in ample time to permit ordering, delivery and documentation and not delay the progress of the work.
- issuing requisitions for all sub-contracted drafting work and shall ensure that all conditions of the Contract are part of such sub-contracts.

415.66 Material Control - All materials for structural components shall be ordered by the Fabricator's Purchasing Manager in full conformance with the material requisitions provided by the Fabricator's Chief Draftsperson.

The Purchase Orders shall contain all information necessary to ensure that materials purchased will comply fully with the terms of the Contract. Where mill certificates and test reports are required, it shall be so stated on the Purchase Order. Instructions shall state when the certificates and reports are to be delivered to the Fabricator.

If a supplier proposes a substitute for any material, the Fabricator's Purchasing Manager shall refer the proposed substitution to the Fabricator's Chief Design Engineer for review. If the substitute is acceptable to the Designer and the Ministry Representative, the Fabricator's Chief Draftsperson shall amend all drawings and requisitions,

withdraw old issues and issue the new versions.

The Fabricator's Receiver shall:

- inspect all materials on arrival for conformance with the Purchase Orders.
- confirm that mill certificates and test reports are provided and that they correctly identify the materials delivered.
- arrange with the Fabricator's Plant Superintendent to store all materials for the Contract in segregated areas.
- ensure all materials are clearly identified with the Contract.
- issue a non-conformance report covering overage, shortage or damage to the materials. Copies of the report shall be provided to the Fabricator's Purchasing Manager and the Fabricator's Quality Control Manager.

The Fabricator's Purchasing Manager shall deliver all documentation to the Fabricator's Quality Control Manager for inclusion in the Fabricator's Quality Control file for the Contract.

415.67 Pre-Pour Inspection and Reporting - The Fabricator's Quality Control Manager shall ensure only documented materials are used for the contract.

Forms used in the casting of members shall be inspected prior to use and shall meet all the requirements of the current CSA-A23.4 and the Design.

Reinforcement shall be fabricated and placed to satisfy the current CSA-A23.4 and the detailed shop drawings for quantity and spacing. Any reinforcement used to anchor cast in hardware by being welded to it will require the Fabricator to be certified to CSA W186-M.

All cast in hardware shall be fabricated and placed according to the current CSA-A23.4. Any hardware requiring welding, including stud welding, will require the fabricator to be certified under CSA W47.1 Division 3 and have approved procedures for each weld.

All pre-tensioning operations shall be done according to the current CSA-A23.4. Care shall be taken to properly execute, supervise and record the operations in a clear and legible manner.

The Fabricator's Quality Control Manager shall file a written report of verification with the Fabricator's General Manager and report any non-conformance that may exist on any of the above items in DBSS 415.67.

415.68 Concrete Control - All materials used in concrete mixes such as cement, fine and coarse aggregates, and admixtures, shall have representative certificates that can verify the quality of them.

Production of concrete shall be done according to the Contract and the Design specifications. Records shall be kept of the mix type and quantity used in each pour.

Handling, placing and vibration of freshly mixed concrete shall be done as per the Contract and the Design specifications. Any variations and interruptions in the process shall be recorded.

The sampling and testing of freshly mixed concrete shall be in accordance with the Contract and the Design specifications and procedures laid out in the current CSA-A23.2. Concrete test cylinders shall be made, cured and tested in accordance with the Contract and the Design specifications.

Reports for all concrete testing shall be kept on file and all non-conformities reported.

415.69 Post Pour Inspection and Reporting - Dimensions of all members shall be verified to ensure tolerances are being maintained as per the Contract and Design specifications.

Verification shall be made of finishes, blockouts, keyways, projecting rebar, sole plates and protection of the cut off strands.

Camber shall be monitored at release and 28 days.

Members shall be stored according to the fabricator's documented procedures.

Members shall only be allowed to be shipped out for erection after they have been approved by both Quality Control and Quality Assurance.

Records shall be kept on file for all items in this section.

415.70 Correction of Non-Conformities - When a non-conformance is encountered, the Fabricator's Quality Control Manager shall determine a recommended disposition and submit such disposition to the Quality Manager and Ministry Representative for acceptance as quickly as possible.

If there is non-conformance to the material specification stipulated in the Purchase Order, the Fabricator's Purchasing Manager shall immediately find out the reasons for the delivery of non-conforming materials. If the material is of a grade superior to that ordered, the Fabricator's Quality Control Manager and the Fabricator's Chief Engineer shall be notified. They shall verify that the material is an

acceptable alternative in all respects. This verification shall include consultation with the Quality Manager and Ministry Representative. If the material is of a grade inferior to that ordered, it shall be rejected and the correct material or a superior material shall be obtained.

If there is non-conformance of material delivery that will delay production, the Fabricator's Contract Administrative Manager and the Fabricator's Plant Superintendent shall be notified immediately. They shall be given revised delivery dates for the adjustment of production scheduling. The Fabricator's General Manager shall determine the alternatives available and shall notify the Quality Manager and the Ministry Representative.

If there are pre-pour non-conformities in form set-up, prestressing, rebar placement, hardware and blockout size and location, the Fabricator's Quality Control Manager and the Fabricator's Structural Supervisor shall immediately review the non-conformance and notify the Fabricator's Chief Design Engineer who may require further investigation prior to submitting corrective action to the Quality Manager and the Ministry Representative for acceptance. If the necessary corrective action will result in delay to production, the Fabricator's General Manager shall be notified for adjustment to the production schedule. The Fabricator's Plant Superintendent shall inform the Fabricator's Quality Control Manager when the corrective actions are being done so that conformance can be verified and the non-conformance report cancelled.

If there are post-pour non-conformities such as honeycombs or large voids after form removal, before de-tensioning the Fabricator's Quality Control Manager shall notify the Fabricator's Chief Design Engineer who will investigate and submit corrective action to the Quality Manager and Ministry Representative for acceptance. Other non-conformities such as low concrete strength, dimensional errors due to faulty forms, variations in camber from design and missing hardware shall all be reported by the Fabricator's Quality Control Manager.

415.71 Welding - All welding on structural components shall conform to the Welding Standards of the Fabricator as Approved by the Canadian Welding Bureau.

All welding consumables shall conform to the Approved Welding Standards and shall be received, stored and conditioned according to the applicable Welding Standards.

All welders and welding operators on the Contract shall be qualified under the requirements of the current CSA W47.1 division 3 or better.

All welding of reinforcing steel, when acceptable to the Designer and Ministry Representative, shall be done by fabricators certified under CSA W186.

415.72 Calibration of Measurement and Test Equipment - All measurement and testing equipment owned by the Fabricator and used in the Quality Control Program shall be calibrated and re-calibrated at the intervals and in the manner stipulated in the current CSA-A23.4.

When the concrete testing services are contracted to a testing firm, then the testing firm shall meet the requirements of the latest edition of CSA A283. The testing firm shall confirm in writing to the Quality Manager and, if requested, to the Ministry Representative that all their measurement and test equipment used in the testing is properly calibrated. If requested, they shall furnish copies of the calibration test data for inclusion in the contract documentation.

415.73 Records Retention - The Fabricator's Record File shall contain the pertinent drawings, purchase orders, bills of material, material mill certificates, test reports, Quality Control documents, and certificates of compliance.

The Fabricator's Record File shall be made available to the Quality Manager and Ministry Representative upon request.

Items in the Fabricator's Record File shall be retained as per company policy.

The Quality Control Manager shall ensure that each file is complete in all respects before it is placed in the Company archives.

415.74 Hold Points - In planning the work flow, it is essential to coordinate with the Quality Manager and Ministry Representative to decide "hold" points for inspection. A list of hold points shall be drawn up by the Fabricator, Quality Manager and the Ministry Representative at a prefabrication meeting. The Quality Manager and Ministry Representative should be continuously informed of progress so that delays are minimized.

The "hold" points will typically include, but not be limited to, the following:

- verification of form size, quality and layout
- after stressing, rebar and hardware placement
- prior to form close up
- sampling and testing of concrete
- release cylinders
- after patching and finishing
- 28 day cylinder results
- shipping arrangements.

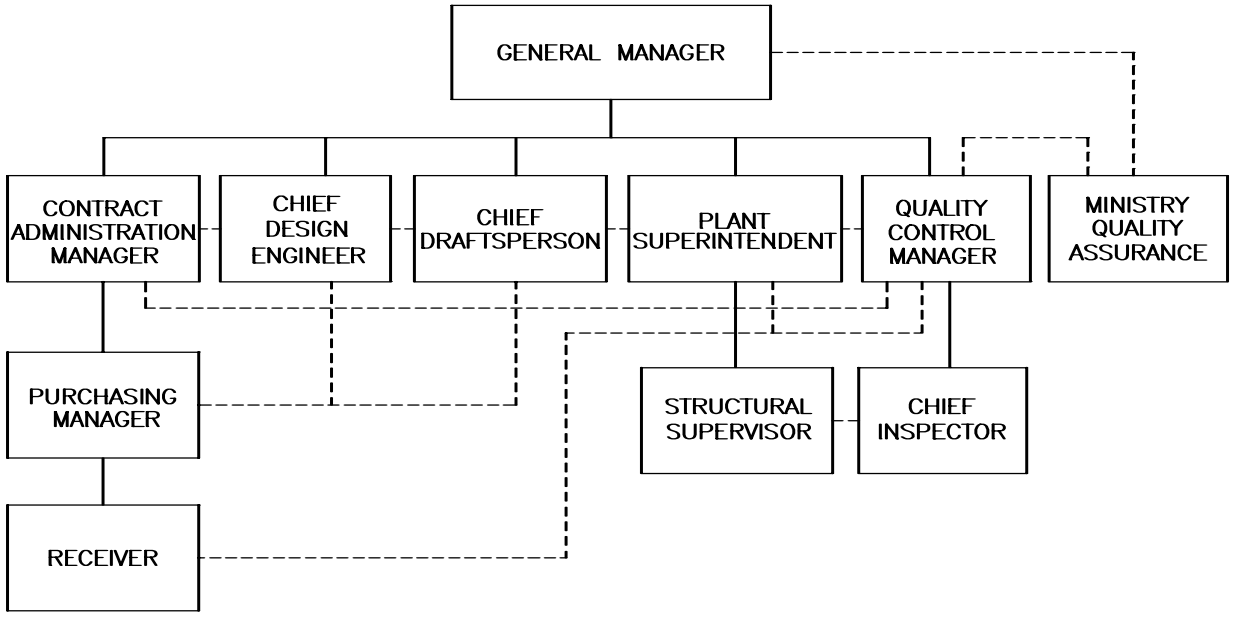
It must be understood that work will not proceed past a

"hold" point until it has been signed off by Quality Control and Quality Assurance. Reports must be completed promptly.

(Refer to the current edition of the PCI Manual - Publication MNL-116 for "Sample Record Forms")

Figure 1

SUGGESTED ORGANIZATION CHART



Direct Functional Reporting _____

Quality Control Reporting -

Signed _____
General Manager

Date _____